InternSync – Final Report

CSC 305 – Software Engineering · Semester 1 – 2026/2027 · Group 3 · Imam Abdulrahman Bin Faisal University
Version 1.0 · Oct 29, 2026 · Milestone M7 (final delivery, Dec 15, 2026) · Author: Salam Alsaour

1. Summary

- **Problem:** co-operative and summer training at universities is run with email, spreadsheets, and paper forms. Students struggle to find suitable training, weekly reports are late or copied, evaluations differ from company to company, and grades and statistics take weeks to assemble.
- **What was built:** InternSync is one bilingual (Arabic and English) web platform for universities, colleges, companies, and students. It covers the whole training cycle, from registration and verification, matching students to vacancies with explanations, and weekly reports and attendance, to evaluations on one rubric, grades with approval, certificates, and statistics.
- **Result:** all 19 product functions of the Proposal were delivered and tested. The measured results meet their targets, except the similarity check on reworded reports and the user acceptance test, which has not been run yet. The system runs at no cost.

2. The System

Aspect	Delivered
Users	Student, company supervisor (and Company Admin), academic supervisor, coordinator, Platform Admin; public certificate check
Functions	F1–F19 of the Proposal: 125 requirements in the SRS (82 Must, 36 Should, 7 Could)
Size	163 API endpoints, 49 database tables, 45 screens, 839 texts in each language
Architecture	React web app (installable, with an offline mode) → FastAPI API with scheduled jobs and an on-server embedding model → PostgreSQL with pgvector; files in private storage (Design, Section 2)
AI	Skill extraction from CVs; matching with a multilingual model (multilingual-e5-small) plus rules; similarity between reports. Every AI result is advisory and explained
Hosting	Zero-cost reference deployment prepared (Vercel, Hugging Face Spaces, Supabase, Gmail); local Docker Compose for development and demonstrations (Deployment)

3. How It Was Built

- **Process:** plan-driven development with incremental delivery (SPMP, Section 1.1). Must functions came first (Increments 1 and 2), then Should (Increment 3), then the Could functions with the testing phase (Increment 4), then the final delivery (Increment 5).
- **Each increment** went through four phases, each with its document in `docs/Increments`:
    1. requirements, with the open decisions and the acceptance tests;
    2. design;
    3. implementation;
    4. testing, with a test report.
- **Quality practices:**
    - every change through a pull request with automated checks (CI);
    - API types generated from the code and compared in CI;
    - a permission test of every endpoint for every role;
    - change requests for every decision that changed a document.

- **Documents:** Markdown is the source of truth. The submitted Word and PDF files are generated from it (document index).

4. Schedule: Planned against Actual

Milestone	Planned	Actual
M1 Requirements baseline	Oct 3	SRS 1.0 on Oct 10
M2 Design complete	Oct 15	Oct 23, 8 days late after CR-05 (one platform for many universities)
M3 Increment 1	Nov 1	Built and tested Oct 24
M4 Increment 2 (all Must)	Nov 15	Built and tested Oct 26
M5 Increment 3 (Should)	Nov 25	Built and tested Oct 27
M6 Testing complete	Dec 10	Could functions and release checks done Oct 28; user acceptance testing not yet run
M7 Final delivery	Dec 15	Documents, guide, demo script, and presentation ready Oct 29

Design took longer than planned because CR-05 widened the scope from one college to many universities and companies. The increments then finished ahead of plan, which left time for all the Could functions and for release checks the plan had not required (browser tests, a load test, a security scan, and held-out AI sets).

5. Changes

ID	Change
CR-01	Zero-cost reference deployment
CR-02	UC-15 added to the use cases
CR-03	Weekly working days in the cycle
CR-04	Change requests logged in the repository
CR-05	One platform for many universities and companies: open registration with verification, Platform Admin, direct vacancy publishing
CR-06	Training type part of eligibility
CR-07 – CR-09	Implementation adjustments of Increments 1–3 (among them the PDF library, files streamed by the API, and sentence-length similarity passages)
CR-10	Offline mode: a saved copy of the session and drafts kept on the device
CR-11	Implementation adjustments of Increment 4 (retention and anonymization, certificate details, the idle limit on the device)

Details and the documents each change affected are in Change_Requests.md.

6. Quality and Results

The numbers below are copied from Key_Results.md, which gives the target and the way to repeat each one.

What	Result	Target
Automated tests	202 backend tests (95% of the code covered), 19 frontend unit tests, 34 browser tests on four engines and sizes	All pass
Permission checks	810 refused calls over 163 endpoints, none allowed	0 (O1)
Matching, relevant vacancy in the top 5	100% on the tuning set and on the held-out set	80% (O2)
Skills found in CVs	97.2%	70%

What	Result	Target
Copied reports flagged	10 of 10 (tuning); 7 of 10 held-out: all copied word for word, none of the reworded	9 of 10 (O5)
Original reports flagged	0 of 10 on both sets	At most 1
50 users at once	95% within 0.21 s, 0 errors	2 s
Grade sheet of 100 students	1.0–1.3 s Excel, 2.4–4.6 s PDF	60 s (O6)
Security scan	0 high-risk findings	0 high
Languages	100% of screens in Arabic and English	100% (O7)
Cost	0	0

7. Requirements Status

From Requirements_Status.md, generated from the SRS, the increment documents, and the tests:

Priority	Requirements	Tested (automated)	Checked (result recorded)	Open
Must	82	54	25	3
Should	36	29	7	0
Could	7	7	0	0
All	**125**	**90**	**32**	**3**

"Tested" counts requirements with an automated test named after their acceptance test; many "Checked" ones are also automated under other names, as their test reports say.

The three open requirements need people or the hosted system:

- **NFR-USA-01:** user acceptance testing.
- **NFR-REL-01:** uptime of the hosted server.
- **NFR-SEC-01:** HTTPS on the hosted system.

8. Risks

Risk (SPMP Section 7)	What happened
Late requirement changes and scope creep	Occurred: CR-05 widened the scope during design. Absorbed by a slower design phase and faster increments; nothing was dropped
Increment 1 grows after CR-05	Occurred and handled as planned: FR-ORG-07 moved to Increment 3
Matching or similarity results unreliable	Partly: matching held on a harder held-out set; similarity misses reworded copies (Section 9). Both stay advisory with a person deciding
Extracting skills from CVs in different formats	A 5 MB CV took 94 s; found by a timed test and reduced to 2–6 s
Arabic in PDF exports	The planned PDF library did not install on Windows; replaced (CR-09), and Arabic is joined and right to left
Permission errors expose data	Did not occur: every endpoint is tested for every role
Free hosting tiers sleep or hit limits	The reference deployment waits for the owner's accounts; demonstrations run on the local stack, as planned
Volunteers for user acceptance testing	Occurred: the test has not been run; the plan, cards, and sheet are ready

9. Limitations and Open Items

- **User acceptance testing (O8)** and the device check on real phones have not been run. The materials are in 05_UAT_Plan.md.
- **The reference deployment** is prepared and off until the owner creates the accounts (Deployment, Section 3). The load test and the security scan were run on the local stack.
- **The similarity check** finds text copied word for word, not reports reworded sentence by sentence (7 of 10 on the held-out set).
- **The AI evaluation sets** are synthetic and written by the same author. The held-out sets are independent of the settings, not of the author.
- **Validation messages for wrongly typed values** (for example letters in a number field) come from the framework in English; the forms prevent most of them, and every business rule answers in the user's language.

10. Lessons Learned

1. **A contract the machine checks prevents integration bugs.** The frontend's API types are generated from the backend and compared in every build, so a mismatch fails the build instead of a screen.
2. **Testing permissions exhaustively is cheap.** One table of endpoints and roles, and one test, checked 810 refusals and caught every route added without a rule.
3. **Measure on data you did not tune on.** The similarity check scored 10 of 10 on its tuning set and 7 of 10 on new data; only the held-out set showed the limit.
4. **Automated scans find what reviews miss.** The security scan showed that the hosted app had no Content-Security-Policy, although the design said it had one.
5. **Check time, not only results.** A timed test found a CV that took 94 s to read, and looking at what the app does while idle found that a session never expired while a window stayed open.
6. **Keep the documents next to the code and change them in the same pull request.** With a change request per decision and a generated requirements status, the documents stayed consistent through 11 changes.
7. **Book people early.** User acceptance testing needs volunteers and dates; it was the only planned activity not completed.

11. Future Work

- Run the user acceptance test and the device check; switch on the reference deployment.
- Detect reworded copies with an embedding-only rule, evaluated on its own held-out set before use.
- Sign-in with the university's accounts (single sign-on) and import of enrolment lists.
- Statistics across universities for the Platform Admin; more languages.

12. Documents

Document	Purpose
SRS	What the system must do (version 1.6)
SPMP	How the project was planned and run
System design and API specification	How the system is built
Increment documents	Requirements, design, and test report of each increment
Change requests	Every change and its effect
Requirements status	Each requirement with its tests

Document	Purpose
Key results	Every measured result
User guide	How each role uses the system
Demo script	The final demonstration
Deployment	The zero-cost deployment and how to switch it on
Final presentation	14 slides with speaker notes; downloadable as PowerPoint or PDF from the deck's page
`Final_Submission/`	PDF of each document, and the SRS in Word and PDF on the course template